

HOUSEHOLD ELECTRICITY EDA PROJECT DOCUMENT

Project Summary

- Developed an end-to-end EDA web application using Python and Streamlit.
- Performed data cleaning, transformation, statistical analysis, and visualization on household electricity consumption data.
- Analyzed electricity consumption patterns across hours, days, and months.
- Created visualizations for voltage, active power, intensity, and sub-metering consumption.
- Analyzed relationships between Global Active Power, Global Intensity, Voltage, and Total Submetering.

Tools & Technology Used

Python, NumPy, Pandas, Matplotlib, Seaborn, Streamlit, VS Code, Jupyter Notebook

Why this project?

The project analyzes household electricity consumption data to understand how electricity usage changes over time and across different consumption categories. The analysis focuses on identifying hourly, weekly, and monthly patterns and understanding relationships among the major electricity measurements.

Dataset Overview

- Records analyzed: 1,048,575
- Original columns: 9
- Date range: 16 December 2006 to 13 December 2008
- Missing values: 4,069 in Sub_metering_3
- Numeric fields were converted using `pd.to_numeric(..., errors='coerce')` before statistical analysis.

Features

- Date – Date of electricity measurement
- Time – Time of electricity measurement
- Global_active_power – Household active power consumption
- Global_reactive_power – Household reactive power consumption
- Voltage – Average voltage measurement
- Global_intensity – Household current intensity
- Sub_metering_1 – Consumption measured by sub-metering 1
- Sub_metering_2 – Consumption measured by sub-metering 2
- Sub_metering_3 – Consumption measured by sub-metering 3
- Total_Submetering – Sum of the three sub-metering categories
- Hour – Hour extracted from Time
- Day – Day of the week extracted from Date

- Month – Month extracted from Date
- Energy_Outside_Submetering – Global Active Power minus Total Submetering

Problem / Questions

- What is the overall distribution of household electricity consumption?
- What are the minimum and maximum values of Global Active Power?
- When did the minimum and maximum power consumption occur?
- How does Global Active Power change over different hours of the day?
- Are there noticeable trends in power consumption over different days of the week?
- How does Total Submetering change over different hours of the day?
- How does voltage fluctuate throughout the day?
- What is the average Global Active Power usage for each month?
- Which sub-metering category has the highest average consumption?
- How does Total Submetering compare with Global Active Power?
- How much energy is consumed outside the three sub-metering categories?
- Are there specific periods where Total Submetering is significantly higher or lower than usual?
- Which hour has the highest Total Submetering on average?

Important Insights

Analysis	Finding
Minimum Global Active Power	0.076 at 19:26:00 on 12 August 2008
Maximum Global Active Power	10.67 at 19:34:00 on 4 March 2007
Average Total Submetering	8.586
Std. deviation of Total Submetering	13.256
Highest average sub-metering category	Sub_metering_3 = 5.934
Total Submetering vs Global Active Power	Correlation = 0.832
Average Energy Outside Submetering	-7.478
Unusual Total Submetering observations	70,494 observations using mean $\pm$ 1 standard deviation
Hour with highest average Total Submetering	08:00, average = 14.408

Additional Observations

- Global Active Power has a mean of 1.108 and a standard deviation of 1.130.
- Global Reactive Power has a mean of 0.118.
- Average Voltage is approximately 239.960.
- Global Intensity has a mean of 4.718.

- The correlation of 0.832 indicates a strong positive linear relationship between Total Submetering and Global Active Power in the analyzed data.
- The negative average value of Energy Outside Submetering results directly from the project's calculation: Global Active Power minus Total Submetering.
- The notebook includes day-of-week and monthly visualizations, but it does not print the numerical highest day or month; therefore those values are not fabricated here.

Source basis: findings above are taken from the calculations and outputs recorded in the supplied HouseholdElectricity.ipynb. No unsupported numerical findings were added.